ASCO Ethical Guidance for the Practical Management of Oncology Drug Shortages

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INTRODUCTION

In early 2023, national shortages of generic chemotherapies used to treat many common cancers forced many hospitals and clinics to allocate drugs in scarcity. Shortages of chemotherapies such as cisplatin and carboplatin are an acute example of an ongoing problem, with professional societies,[1] media, policy-makers, regulators, and the Biden administration[2] calling for action and relief. Shortages that affect patients with cancer are not unique to antineoplastic therapies; they have also been subject to clinically significant shortages of pressors,[3] benzodiazepines,[4] and antibiotics,[5] among others. Given the complexity of the problem and the long timeline toward solutions addressing supply chain, demand, and manufacturing challenges, chemotherapy shortages are likely to continue into the foreseeable future. Patients and oncologists will be faced with delays, waitlists, or suboptimal clinical options and outcomes.

Resource shortages in oncology and medicine generally pose ethical issues for clinicians and their patients. Promoting values such as trust, resource stewardship, transparency, and equity is critical.[6] The risk of increased cancer recurrence and death and the potential harms involved in therapeutic alternatives during severe shortages[7] force clinicians and health care systems to decide which patients receive scarce medications and which do not, that is, explicit rationing. In this context, evidence-based care must also be maintained to the greatest extent possible. Insufficient support for developing and implementing fair rationing strategies to meet agreed-upon goals can perpetuate inequities in care, contribute to clinician distress, and lead to conflicts between and among clinicians and patients.

To aid clinicians, ASCO has issued clinical recommendations and ethical principles for the development and implementation of allocation plans.[8] This article expands on ASCO's guidance[9] to move toward practical strategies for the ethical management of shortages. We begin with an overview of ethical frameworks for drug shortages to educate and inform oncologists. Then, we highlight the process and principles of multistakeholder deliberation in allocation planning and enactment. We conclude by touching on practical considerations for blending process strategies with allocation principles to develop a functional plan for oncology practices.

ASCO's disease-site specific guidance documents can aid in technical implementation via evidence-based clinical approaches, with this document serving as a complement to the guidance provided therein. Planning ahead and developing a clear allocation strategy can obviate situations where individual oncologists are forced to choose between their patients (bedside rationing). It can also protect against bias and monitor for inequities. Specific allocation plans and priorities are likely to vary on the basis of the setting, the severity of the shortage, and the scarce drugs in question. Given this, the ethical and practical considerations sketched here do not constitute legal or medical determinations, nor do they imply statements of the standard of care.

Ethical Considerations and Framework Approaches

Health care resource shortages occur frequently in oncology[10,11] and across medicine. During the height of the COVID-19 pandemic, there were concerns about critical care resource and personnel capacity[12,13] along with shortages of COVID-19 vaccines and therapeutics.[14] Recent shortages of intravenous contrast and cytarabine have also affected cancer care in the United States,[15,16] and other shortages have affected patients with cancer of all ages and across the globe.[17] Resource shortages pose a myriad of ethical issues,[18-21] forcing clinicians to decide which patients receive scarce drugs and which do not. General bioethical principles are helpful for providing an overall ethical frame for allocation during resource shortages.[20] These principles include beneficence (promoting individual patient well-being), nonmaleficence (avoiding unnecessary harm), stewardship (careful management and responsibility for scarce resources), trust (maintaining transparency and communication), respect for autonomy (respecting an individual's right to make their own health care decisions), and distributive justice (fair distribution of burdens and benefits across a population). These principles, however, cannot help clinicians and institutions make ethical allocation decisions or direct an ethically grounded allocation process. Below, we describe specific ethical considerations related to the process and allocation of scarce resources (grounded in clinical science and principles in bioethics like those outlined above) to help guide clinicians and institutions in these challenging contexts. Notably, allocation plans should be informed by a cancer care institution's carefully considered core values.

Process principles help direct an allocation process that promotes fair decision making and prevents arbitrary or biased distribution practices. They also promote meaningful opportunities for stakeholder participation as an allocation system is established. A focus on process also recognizes that deliberation may not yield a unified consensus on who should be prioritized. Even in the absence of consensus, process principles help support decisions made through consultative participation. In support of these aims, ASCO has highlighted the process principles of *Stakeholder Participation*; *Consistency and Review*; *Relevance to the Circumstances*; *Equity*; and *Transparency*, *Trust, and Communication*. Definitions and implementation strategies for each principle are shown in Table 1.

These principles are based on the accountability for reasonableness (A4R) framework originally proposed by Daniels and Sabin,[22] which has been adopted for drug shortage management by Valgus et al,[23] Rosoff,[24] and others in oncology.[25-27] The A4R framework also served, in part, as the foundation for previous ASCO guidelines on the allocation of scarce resources.[28] Process principles such as *Stakeholder Participation* and *Equity* are particularly significant for developing equitable allocation plans. Their application can help reduce inequities in scarce resource distribution by ensuring that allocation strategies incorporate and meet the needs of patients from disadvantaged groups and those most affected by allocation decisions.

Together, these broad ethical principles and process principles inform specific allocation principles, which identify ethically justifiable reasons for prioritizing a group of patients to receive scarce chemotherapy when there is not enough for those who might benefit from it.[14] In Table 2, ASCO has highlighted nine such allocation principles. For example, the *Lives Saved* principle seeks to maximize the number of patients who survive after receiving the scarce drug.[18,27] One way this may be practically implemented is to prioritize patients receiving chemotherapy for curative intent over those treated with palliative intent. Overall, allocation principles are important for promoting justice and reducing bias in allocation decisions.

How to Use Process and Allocation Principles to Prepare for and Respond to Shortages

To minimize inequities and bedside rationing, allocation should take place at the highest level possible (ie, furthest from the bedside) and through a committee of stakeholders with relevant expertise, when feasible. In most cases, the institution, or possibly the health system, is the highest level at which allocation can feasibly occur. Committees should include multidisciplinary stakeholders representative of the organization within which allocation occurs. Committee stakeholders should include, but are not limited to, clinicians, pharmacists, nurses, ethicists, patients, administrators, and community members. At small hospitals and practices, individuals may need to act in multiple capacities; institutions without ethics expertise are encouraged to seek guidance from national organizations such as ASCO. It is critical that technical experts, with experience in managing the allocation and delivery of scarce care, participate in the development of the allocation plan. Clinicians with expertise in the scarce therapeutics play an important role in stakeholder deliberation. They help other stakeholders understand the medical consequences of potential allocation plans and their medical benefit.

Process principles such as *Equity* and *Relevance to the Circumstances* can be implemented in various ways. *Equity*, for example, may be implemented by including patients from disadvantaged groups in multistakeholder committees and by factoring in their concerns and insights during allocation planning. Suggested implementation of this principle also includes recording the distribution and outcomes of the scarce drug. This will assess if the initially agreed-upon strategy is working as intended and help determine if any groups are inadvertently and disproportionately subject to worse outcomes because of shortages. Committee members can begin identifying and addressing barriers to next-best alternatives that might doubly affect deprioritized groups, such as an increased financial burden for those who need to be treated with an alternative. Another process principle to consider in maintaining an ethical deliberative process is *Relevance to the Circumstances*. This principle may be implemented by clearly articulating the values guiding allocation decisions and by developing clear allocation goals that can be tracked and measured.

TABLE 1. Ethical Process Principles and Implementation Strategies for Drug Shortages

Ethical Process Principle	Implementation Strategies
Stakeholder participation: All relevant stakeholders, including those most affected by allocation decisions, should be involved in allocation planning	Multidisciplinary committees should include clinical staff, patients, pharmacists, infusion nurses, clinical ethicists or ethics committee members, and other stakeholders Committees that are unable to include all stakeholders should consult with affected individuals and groups as part of the development process Oncologists without practice or institutional support can consider joining with other practices in their state or region to develop a strategy
Relevance to the circumstances: All relevant information—including evidence-based data, relevant patient population information, organizational goals, and values—should figure in allocation planning	Articulate relevant values and principles to guide allocation decisions Develop clearly stated goals for an allocation plan that can be tracked and measured during planned reviews Strategies to conserve drug and maintain the standard of care are implemented first Primary allocation: Incorporate evidence-based clinical guidance; consider factors such as likelihood of benefit and certainty of benefit, existence of effective alternatives, relative toxicity, and phase of treatment Secondary allocation: Fair and unbiased process for deciding between patients with the same position after primary clinical analysis Create or follow a framework grounded in organizational values and responsive to the patient population Strive to create mitigation and allocation strategies as explicitly as possible so that individual physicians are not forced to make allocation choices and to limit heterogeneity in decisions
Consistency and review: Allocation plans should be applied consistently; committees should convene regularly to review policy when necessary and address allocation issues as they arise	Apply allocation policies consistently Include a timely and consistent appeal process Collect data on the effectiveness of the policy at achieving the stated allocation goals and, if possible, patient outcomes Perform data reviews at regular intervals, ideally aligned with key timepoints (eg, new supply estimates), to assess concordance with goals and for unanticipated issues (ie, inequities) Revise policies, as needed, on the basis of findings from the above Maintain the committee to monitor for shortages and plan for future events
Equity: The voices of patients from historically marginalized groups should be included in allocation planning, and allocation strategies should be sensitive to the needs of such groups	Include patients from disadvantaged groups fairly in allocation decisions, and explicitly consider whether allocation frameworks might unintentionally continue or worsen existing inequities Consider whether patients from disadvantaged groups should receive priority in allocation decisions (eg, if a given minoritized group is known to have worse outcomes from the disease treated by the drug in short supply or is more likely to present with more advanced disease) Develop decision procedures that avoid awareness of the specific patient in need to prevent bias and/or preferential treatment for powerful patients and clinicians Facilitate access to care (eg, transportation, childcare, disability accommodations) so that access issues are not a factor in allocation of drugs in shortage Share drugs, if possible, and exchange information about shortages between institutions and networks of institutions; avoid hoarding and stockpiling; ASCO opposes hoarding and stockpiling of chemotherapy medications Promote access for patients in remote and underserved geographic locations, and mitigate impact on economically disadvantaged patients Ensure that patients and their families who cannot receive the scarce medication receive the best available treatment. This includes palliation and psychosocial support
Transparency, trust, and communication: Drug shortages and allocation plans should be clearly communicated with hospital staff and affected patients and families	Communicate clearly with patients, families, and members of the care team Include language translation and interpretation in communications plans Inform patients if shortages could affect care Whenever possible, clearly and publicly describe any mitigation and allocation plans that will be implemented Consider targeted guidance to not unnecessarily worry patients and families who will not be affected

Allocation principles support scarce drugs being distributed ethically across patient populations. These principles and how each might be practically applied are listed in Table 2 along with their potential benefits and shortcomings. Allocation principles can be applied either individually or in conjunction with others. This is because clinicians may find that frameworks on the basis of a single principle are insufficient for promoting just distribution practices and/or avoiding bias. For example, using only the *Lives Saved* principle would prioritize those with a high rate of cure when receiving the scarce therapy, even if they could also be cured with another therapy in ample supply. By itself, this principle would deprioritize a group with a slightly lower cure rate but without other alternatives, which many would consider

TABLE 2. Allocation Principles and Their Respective Benefits and Limitations[29]

Allocation Principle	Practical Method of Application	Practical Benefits/Justification	Practical Issues/Limitations
Lives saved, Life-years saved[a]	Prioritize by curative intent	Promotes a utilitarian concept, aligns with evidence base Easy to apply in complex situations when shortage affects multiple diseases and regimens	Does not equal likelihood of long-term survival, can have non–curative-intent patients with one disease living longer than curative intent with another disease Does not account for additive value of one drug in a multidrug regimen Noncurative deprioritization may exacerbate disparities as persons with less access to health care more often present in later stages Requires consideration of whether age should be a consideration in allocation decisions
	Prioritize by expectation of long-term survival	Promotes a utilitarian concept and aligns with evidence base Can be useful when deciding between a small group of patients with high levels of evidence for regimens	Requires cross-trial and cross-disease comparison with differing levels of evidence Can be difficult to enact as a strategy if there are many patients and regimens across many disease types
	Prioritize by either of the above but account for lowest usage per patient	Useful to maximize drug across a heterogeneous group Aligns with youngest first in most cases The research consideration of double denial discussed under reciprocity also applies here	Practically difficult to align with the above in a timely manner
Worst off	Prioritize those with inferior or medically contraindicated alternatives	Useful for determining the need on the basis of lack of effective alternatives Reasonable metrics include differences in overall survival (or established surrogate outcomes)	Worst off defined as "those closest to death" would prioritize those least likely to benefit and is not an appropriate definition in the context of chemotherapy shortages With many diseases and alternatives, difficult to stratify by individual regimens/alternatives; a binary or simplified metric may be practically acceptable (ie, is the alternative statistically inferior? [yes/no], or is the alternative statistically inferior and decreases the HR for long-term survival by xx%? [yes/no])
Youngest first	Prioritize children and younger adults	Directs treatment to a group that is generally recognized as worth prioritizing Often aligns with expectation of long-term survival and lowest usage per patient	If enacted alone, can prioritize younger patients with low likelihood of long-term survival over older patients with higher likelihood of long-term survival If applied in groupings (eg, <18 years, 18-30 years, etc), it is difficult to justify different priority between individuals who are near cutoffs Prioritization with a diminishing multiplier (ie, a continuous score for priority that decreases by age) may be difficult to enact in a multiple criteria system and may be used as a tie breaker
Lottery	Random choice	Useful when prioritizing within a homogenous group	If applied by itself does not account for known differences in efficacy, toxicity, and alternatives. Best use may be as a tie breaker within groups
First-come, first-served	Waitlist on the basis of time from the known need for scarce medication	Useful when prioritizing within a homogenous group	Favors the well-connected, affluent, etc Can compound disparities
Reciprocity	Prioritize a select group of individuals who have previously provided some sort of usefulness, good, or sacrifice (eg, those patients who are participating on a clinical trial where the scarce drug is given as a standard of care)	Recognizes altruism of patients and importance of advancing research Allows patients to not be denied scarce drugs via standard treatment allocation and study allocation If study completion/fidelity would be compromised, this also recognizes the participation of prior patients who have completed treatment	Can be coercive if patients are prioritized only because of trial participation Can exacerbate inequities by giving preference to patients who are offered clinical trials and can participate

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TABLE 2. Allocation Principles and Their Respective Benefits and Limitations[29] (continued)

Allocation Principle	Practical Method of Application	Practical Benefits/Justification	Practical Issues/Limitations
Instrumental value	Prioritize on the basis of future usefulness	Allows for care to those who are needed for ongoing care delivery in a crisis, such as first responders	Little relevance for cancer drug scarcity as this is value laden
Duty to care for existing patients within the home institution	Prioritize patients who are currently being treated (not synonymous with first-come first-serve as it does not differentiate on the basis of the start date)	Allows for evidence-based completion of regimens that have already been started	Unless used in conjunction with utilitarian goal, can prioritize patients with little benefit
	Limit transfers of care specifically for scarce chemotherapy (not new diagnoses/ progressions)	Allows for understanding of supply and demand in a consistent manner such that patients who are prioritized can receive the scarce drug Dissuades advantaged patients who can travel from accessing other supplies Promotes consistency across institutions	Limits the effectiveness of sharing across institutions or regions Can prioritize advantaged institutions (those with more drug) Can exacerbate inequities

NOTE. Below are a series of allocation principles that are commonly used when developing plans for allocation of drugs or other resources in short supply. Of note, most allocation plans use several of these principles together, in either a tiered or combined approach. Recognition of these principles (and their benefits and limitations) can help institutions to develop a rational, practical policy for allocation of scarce drugs.
Abbreviation: HR, hazard ratio.
[a]Notably, the difference between lives saved and life-years saved depends on whether age of patient (and how many years they are expected to live) is deemed ethically relevant.

unjust. Clinicians might therefore adopt a multiprinciple framework that prioritizes several groups at once, like those with the greatest medical need *as well as* those with better odds of long-term survival.[29] Multiprinciple frameworks can also prioritize groups sequentially (eg, first providing access to those with the greatest need and then making further allocation decisions according to survival). Clinicians must decide which principles are appropriate on the basis of the needs, resources, and practice contexts of their respective institutions.

Practical Considerations in Creating Process Strategies and Developing Allocation Criteria

Practical considerations involved when developing and implementing an allocation plan are likely to depend on contextual factors such as practice size, resource availability (including availability of staff support and both the urgency and severity of the scarcity), and previous scarcity planning.

Convening Stakeholders

Practices may face issues in convening stakeholders for deliberation, because of the difficulty in identifying individuals with relevant expertise, concerns over legal liability, and the potentially significant uncompensated time and effort required to create and maintain an allocation management plan. In addition to clinicians, relevant stakeholders include those internal and external to the institution or practice, such as oncologists, nurses, physician assistants, pharmacists and pharmacy buyers, clinical pharmacists, ethicists and/or members of hospital ethics committees, social workers, patient advocates, clinical researchers and research personnel, communications personnel, risk managers and legal counsel, hospital administrators, payors, spiritual care leaders, and electronic health record programers. Not all stakeholder deliberations, however, will necessarily require (or allow for) involvement of all such parties.

Each committee member can play a unique role within the committee, drawing from their unique skills and experiences: social workers can help identify mental health resources for patients affected by shortages; clinical ethicists can identify ethical issues that arise in the context of ethics consultations and shortages; pharmacists can assess how to track supply and demand of scarce resources and update committee members on availability and estimated future supply. Where convening multiple stakeholders in a small or lower-resourced setting proves to be difficult, a single stakeholder could serve multiple roles within the committee. For example, social workers trained in clinical ethics may serve both roles. In addition, clinicians might form teams that include practices outside of their system, with community health workers and advocates or with local or national professional organizations.

Internal stakeholder committees will benefit from leaders with content expertise, seniority, and organizational connections to ensure that the process principles are maintained, recognized, executed, and enforced across the organization. Committees can be charged with developing and revising policies, highlighting policy updates, monitoring shortages, communicating decisions and allocation processes, and performing data reviews. Within health care organizations, committee members can liaise with their departments and clinical/operational colleagues, informing them of regular updates from committee meetings. Committees should convene until the shortage is resolved, meaning that there are no restrictions on the use of the scarce drug(s), a plan is in place to monitor for recurrence, and all follow-up data analyses are completed.

Communication

Consistent, clear external and internal communication promotes the process principle of *Transparency, Trust, and Communication* (Table 1). Internal communications can be bolstered through regular meetings to assess and document the supply and demand of the scarce drug(s), discuss the function of the plans as developed, and identify emerging issues. Technical support for communication includes the institution's ability to maintain secure, centralized documents for allocation plans, drug distribution, and affected patients. Secure documentation can help to reduce fractured communication and minimize information gaps in planning and decision making. Updates may include information about drug availability, allocation strategies, and resources for discussing drug shortages with patients such as standardized talking points to reduce inaccuracies. Clear communication to clinicians about when allocation management begins and ends, and for which patients, is also essential to minimize inequities.

When communicating with patients, transparency must be balanced with awareness of potential to provoke patient anxiety and psychological distress. Mass internal and external communication about drug shortages should not replace nuanced patient-clinician conversations. Patients should be provided with information that relates to their specific course of care and the institution's capacity for treatment options. Establishing and supporting transparent communication from the institution and in individual clinician encounters promotes shared decision making and goal-concordant care, wherever a patient may fall in the allocation strategy. Patients affected by allocation decisions should be given additional resources as available, such as supportive care and patient advocate/ombudsman information.

Pharmacy and Drug Availability Updates

Maintaining and promoting relevant communication among committee members requires up-to-date information about the status of drug shortages and promotes the *Relevance to the Circumstances* process principle. Pharmacists and/or pharmacy administration should serve on the committee and can provide updates about available drug supply, identify existing supply and drug expiration dates, and communicate with suppliers to provide the committee with estimates about scarce drug amounts, delivery dates, and drug availability timelines. They can also collect data to track and estimate patient demand, identify which patients are currently receiving scarce drugs, and work with clinical teams to confirm allocation, who should be treated with alternatives and what the logistics of safely switching to alternatives are. Finally, these lists can be used to assess when allocation can be expanded or stopped altogether and identify which affected patients should be switched to the previously scarce drug. All these processes are also based on and uphold the principles of *Equity* and *Consistency and Review*.

Practical Considerations in Developing Allocation Criteria

Balancing Specificity With Implementation

Lives Saved, *Life-Years Saved*, and *Worst Off* are frequently included as allocation principles. When they are used and a drug in short supply affects only a small number of cancer types or is only used in a small number of treatment regimens with similar levels of evidence, prioritization through review and comparison between affected regimens and alternatives is reasonable. This is also the case if these principles are used and a shortage occurs at a smaller practice where only a small number of patients are affected.

However, at larger practices and/or when a shortage affects many individuals with different cancers and regimens, creating regimen-by-regimen prioritizations can require many cross-trial and cross-disease comparisons with varying levels of evidence and comparators. In these cases, it may not be feasible or timely to assess the large number of regimens and alternatives head-to-head. Furthermore, there is inherent uncertainty in many such comparisons, and there is a substantial variation between individuals on the basis of prognostic risk beyond just what treatment is selected. Thus, prioritization on the basis of individual treatment regimens and alternatives may not in fact adhere to the underlying distributive justice principles better than more general delineations, such as prioritizing by curative over palliative intent. Beyond the issues of inefficiency, confounding, and bias that such general delineations can help overcome, additional benefits include transparency and broad stakeholder understanding of the delineation and its justification (patients and clinicians should be aware of therapeutic intent and that potential cure is good worth prioritizing). Even such a delineation, however, is imperfect. For instance, treating with curative intent does not necessarily equate to long-term survival. A patient with a slow-growing neoplasm receiving palliative treatment could live longer than a patient receiving curative-intent treatment for a cancer with high risk of recurrence and death. Similarly, those receiving palliative-intent treatment may derive greater benefit from a scarce drug than a patient with a favorable prognosis with or without curative-intent adjuvant therapy. Treatment intent is discussed in more detail below.

Matching Prioritization With Demand

Categories of prioritization should be based on agreed-upon goals and allocation principles so that categorizations accurately represent patients affected by, and at risk for, allocation. Without matching categories to expected demand, many patients may fall into one priority group, making it difficult to prioritize between them. In situations where allocation decisions must be made within a homogenous group of patients, allocation principles that treat people equally (eg, lottery) can be useful.

Patient Discharge and Transfer

Regional health care systems and organizations are encouraged to discuss drug availability across the region and coordinate patient discharge and transfer strategies. Because of federal regulations, it may be more feasible to transfer patients than to send supplies of drug from one institution to another. Multidisciplinary committees can standardize referrals to minimize inequities with respect to who can obtain treatment at the referral center. This will also allow the referral center to estimate the number of new patients accurately. Transfers of existing patients for the purpose of obtaining scarce therapy are difficult because they cause an adversarial situation between referring and referral centers, both of which are trying to adhere to duty to care for their patients. Larger centers that might have more supply on hand are encouraged to work with smaller referring practices to maintain equity for all those who live in their service areas.

Allocation in Research

Prioritizing patients treated with scarce chemotherapy treatments during a research study is a complex matter, and a full exploration of this issue is beyond the scope of these guidelines. Nonetheless, given the importance and prevalence of clinical trials in oncology, ASCO provides the following guidance regarding scarce drug allocation decisions involving drugs currently part of clinical trials. Reasons against prioritizing research participants include known inequities in trial enrollment, which may precipitate inequitable allocation, and prioritizing use of a scarce drug without proven efficacy, which may violate the *Lives Saved* and *Life-Years Saved* principles. If research participants are considered for prioritization, there are several strategies for mitigiating these issues. These include limiting the prioritization of research participants to instances in which the scarce drug is part of a standard of care backbone or a comparator arm; when there are no substitutions allowed and all efforts to incorporate a substitution have been made; or as a tiebreaker among otherwise equally prioritized patient groups. If chemotherapy treatments are in short supply, new studies using such treatments should generally not be opened unless there are exceptional circumstances. If there are studies supplying their own scarce chemotherapy, treatment may be continued and not be considered part of the allocation process as these are outside the normal supply chain.

Curative Versus Palliative Intent

Oncologists might have difficulty in defining curative intent when cure or long-term remissions are issues of probability rather than certainty.[30] In the modern era, where prolonged disease control is attainable without cure, such distinctions have become even less certain and potentially less clinically meaningful. Allocation plans that prioritize by curative intent, however, could remain justifiable when evidence suggests that an indication for treatment is (1) life-threatening without the therapy and (2) likely to be cured with therapy. In such cases, prioritizing treatments with curative intent reflects the *Worst Off* and *Lives and Life-Years Saved* principles. Patients and oncologists have identified intent to cure as important during allocation decisions.[31] Nevertheless, its use should be considered alongside other important allocation principles, with the understanding that months added to the end of life are of substantial value to patients and their clinicians.

In addition, the shortcomings of prioritization on the basis of curative intent versus palliation are most evident when there is marginal benefit for a scarce drug in the curative setting and substantial benefit in the palliative setting. For example, if a scarce drug used in the adjuvant setting boosts an already high chance of cure only slightly, but it can add years of disease control in the palliative setting, it is less clear that curative intent should be the basis for priority. In this case, prioritization by curative intent does not allocate the scarce drug to an indication that is life-threatening without the therapy. Here, curative intent poorly reflects the principle of prioritizing *Worst Off.* Moreover, palliative therapy strongly promotes the principle of *Life-Years Saved*, recognizing the goals of disease palliation and life extension. Context, as described elsewhere in this guidance, matters greatly.

In conclusion, the 2023 shortages of carboplatin and cisplatin serve as a reminder that oncology drug shortages should be anticipated and require planning. This means not only establishing allocation principles for explicit rationing but also developing process principles to ensure that allocation principles are created in an ethical manner. This will help oncologists and institutions maintain high-quality patient care and prepare for potential shortages along with their downstream consequences. Familiarity with the management of drug shortages and the principles of allocation should be part of medical training and continuing education efforts. Institutions should have a standing committee and process plan, if not an allocation plan, in place ready to address new shortages. Individual oncologists are obligated to educate themselves on their institutions' process to communicate with patients, remain up to date on evidence-based regimens and alternatives, and reflect on when they might need support when talking to patients; they should also participate in allocation plan development when possible. Collectively, the oncology community has an obligation to anticipate future drug shortages and to be prepared to address them through thoughtful, fair, evidence-based, and efficient allocation and to involve patients and other stakeholders in these efforts. By doing this, we can work to ensure the best outcomes possible while adhering to robust ethical principles.

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ASCO Ethical Guidance for the Practical Management of Oncology Drug Shortages

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